

MUHAMMAD ASAD

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PROFESSIONAL SUMMARY

Second-year Computer Software Engineering student and Backend Development Intern at ByteCorp Technologies, building a production-style Django REST Framework job board backend with JWT authentication and role-based access control. 15+ open-source projects across Python/Django, Java, C#.NET, and JavaScript, including an AI-powered resume analysis platform (ResumeAI), with strong foundations in Data Structures & Algorithms, OOP design, and software architecture patterns.

EDUCATION

Bachelor of Engineering in Computer Software Engineering - Sep 2024 - Sep 2028

Bahria University - Karachi Campus

Intermediate - Computer Science (FSc) - Completed

Malir Cantt College

EXPERIENCE

Backend Development Intern (Hybrid) - Jul 2026 - Present (3-Month Program)

ByteCorp Technologies - Training Program

- Collaborating with a technical mentor on the end-to-end development of a production-style backend system, from database design through API implementation and security.
- Delivered Phase 1 on schedule: REST API implementation with JWT authentication (15-min access / 7-day refresh tokens) and role-based access control across 3 roles, following a structured PR-based workflow with mentor review.
- Taking ownership of independent modules, including schema design, authentication, and authorization, while adhering to industry-standard Git and version control practices.

Java Development Intern (Remote) - 2026

AICTE Oasis Infobyte Internship Program (OIBSIP) - Java Development Track

- Completing a 5-task Java internship track; built an ATM Interface (Java OOP, ArrayList) and an Online Examination System (Java Swing, timer-based test flow).

Software Development Intern (Remote) - 2024 (4-Week Engagement)

CodeAlpha - Project-Based Internship

- Delivered two production-quality Java applications (Student Grade Manager, Hotel Reservation System) under real internship specifications, both reviewed on GitHub.
- Applied clean code principles, OOP design patterns, Git version control, and project documentation throughout the engagement.
- Both submissions were adopted as reference implementations on the internship platform following positive architecture review.

PROJECTS

ResumeAI | FastAPI, Next.js 15, TypeScript, PostgreSQL, JWT, OpenAI/OpenRouter API, Docker | GitHub

- Built a production-ready AI resume analysis platform with ATS scoring, skill-gap analysis, and job-description matching; 18+ REST API endpoints, JWT authentication, and Docker Compose deployment.

FoodHeaven - Restaurant POS System | Python, Django REST Framework, PostgreSQL, JavaScript, PyWebView | GitHub

- Built a native desktop Point of Sale system for restaurants with a dark-themed register UI, real-time cart management, GST calculation, and a Kitchen Monitor Display for live order tracking.
- Developed an Analytics Dashboard using Chart.js for sales trends, along with printable invoice generation and multi-mode checkout backed by a PostgreSQL database across modular Django apps.

Job Ranking System (SkillMesh) | Python, DSA, Greedy Algorithms, OOP | GitHub

- Built a DSA-based candidate ranking and recommendation engine that scores applicants using skill matching, experience scoring, and greedy budget optimization algorithms.
- Implemented a Max Heap for efficient top-candidate retrieval, with an Explain Mode for step-by-step algorithm tracing.

Requirement Genie | Python, NLP, AI/ML, Streamlit | GitHub

- Developed an AI-driven requirements engineering assistant that automates generation of SRS (Software Requirements Specification) documents using NLP techniques.
- Integrated a conversational interface for systematic requirements gathering, reducing manual documentation effort.

Algo Puzzle Board | HTML, CSS, JavaScript, Data Structures & Algorithms | GitHub

- Engineered an interactive algorithm visualization platform covering sorting, pathfinding, N-Queens, Knight's Tour, Graph Coloring, TSP, Huffman Coding, Dijkstra's, and Kruskal's MST.
- Designed a modular dashboard architecture with animated step-by-step algorithm walkthroughs to aid conceptual understanding.

Food Heaven Website | HTML, CSS, JavaScript, Bootstrap 5, .NET Framework | GitHub

- Built a restaurant management website with a modern UI, dynamic menu display, and a table reservation system with responsive design.
- Implemented client-side interactivity using vanilla JavaScript and Bootstrap 5 grid for a seamless experience across devices.

KernelFlow - CPU Scheduling Simulator | React.js (Vite), Tailwind CSS, FastAPI, Python, Docker | GitHub

- Built a full-stack OS CPU scheduling simulator implementing 6 industry-standard algorithms (FCFS, SJF, SRTF, Round Robin, Priority preemptive/non-preemptive) with a React/Vite glassmorphism UI and a FastAPI backend.
- Designed an interactive tick-by-tick Gantt chart, live ready-queue visualizer, and analytics dashboard (avg. waiting time, turnaround time, CPU utilization, throughput), plus side-by-side algorithm comparison and a recommendation engine; containerized with Docker Compose.

Additional projects (Hotel Reservation System, Decode & Discover, Streamlit Dashboard Suite, and 10+ Python/Java mini-projects) available on GitHub and Portfolio.

TECHNICAL SKILLS

Languages: C#, Java, Python, JavaScript, MIPS Assembly, SQL, HTML, CSS

Frameworks & Libraries: Django, Django REST Framework, Bootstrap 5, Streamlit, Pandas, Matplotlib, NumPy, Java Swing, C# Windows Forms, PyWebView, Chart.js

Databases: PostgreSQL, MySQL, SQL Server, SQLite

APIs & Authentication: REST APIs, FastAPI, Django REST Framework, JWT Authentication, Role-Based Access Control (RBAC), Ownership-Based Authorization

Developer Tools: Git, GitHub, VS Code, Visual Studio, IntelliJ IDEA, PyCharm, Netlify

Core Concepts: OOP, Data Structures & Algorithms, Design and Analysis of Algorithms, REST APIs, NLP, Computer Architecture, Agile Basics

Software Architecture & Design: Design Patterns, Design Principles (SOLID), MVC, Client-Server, Layered Architecture, Peer-to-Peer, Pipe-and-Filter

ACHIEVEMENTS & HIGHLIGHTS

- 15+ open-source projects on GitHub across Java, Python, C#, and JavaScript with a consistent commit history demonstrating real-world development discipline.
- Currently completing a Backend Development Internship at ByteCorp Technologies and a Java Development Internship under the AICTE OIBSIP program (both ongoing).
- Built ResumeAI, a full-stack AI resume analysis platform (FastAPI + Next.js) with LLM-powered ATS scoring and job-matching, deployed via Docker.
- Portfolio deployed on Netlify showcasing 10+ production-quality projects spanning web development, AI tools, algorithm visualizations, and systems programming.
- CodeAlpha Internship: completed a remote software development internship, delivering enterprise-style Java applications under real project deadlines.
- Self-directed learning: independently mastered Streamlit, Pandas, NLP basics, and data visualization tools beyond coursework; shipped multiple deployable applications.
- Built an AI-powered requirements engineering tool (Requirement Genie), demonstrating initiative in applying NLP to real software engineering problems.